

# Customer Churn Prediction & Retention Analysis

A Random Forest-based churn model built on the IBM Telco Customer Churn dataset, translating 7,000+ customer records into a targeted, revenue-protecting retention strategy.

**78.6%**

MODEL ACCURACY

**0.824**

ROC-AUC SCORE

**1,382**

HIGH-RISK CUSTOMERS

**26.4%**

OVERALL CHURN RATE

## Table of Contents

|    |                                                  |    |
|----|--------------------------------------------------|----|
| 01 | Executive Summary                                | 3  |
| 02 | Business Problem & Objectives                    | 3  |
| 03 | Tools & Technical Stack                          | 4  |
| 04 | Dataset Overview & Data Cleaning                 | 4  |
| 05 | Exploratory Data Analysis (Chart-by-Chart)       | 5  |
| 06 | Feature Engineering                              | 8  |
| 07 | Predictive Model: Random Forest Classifier       | 8  |
| 08 | Model Evaluation: ROC Curve & Feature Importance | 9  |
| 09 | Risk Segmentation & Retention Playbook           | 10 |
| 10 | Business Impact & Conclusion                     | 11 |

## 1. Executive Summary

Customer churn is one of the most expensive problems a subscription business can have — every customer who leaves takes their lifetime value with them, and replacing them costs far more than keeping them. This project analyzes the **IBM Telco Customer Churn dataset** (7,043 customers) to understand *why* customers leave and to build a model that flags *who* is likely to leave next, early enough for a retention team to act.

The workflow follows a complete analytics lifecycle: data cleaning and validation, exploratory analysis to surface behavioral patterns, feature engineering to encode business logic into the dataset, and a Random Forest classifier to turn those patterns into a scored prediction. The result is not just a model — it is a decision-support tool: a ranked list of at-risk customers, the reasons they are at risk, and a segment-specific playbook for what to do about it.

**78.6%**

ACCURACY

**0.824**

ROC-AUC

**1,382**

HIGH-RISK FLAGGED

**26.4%**

CHURN RATE

**Headline finding:** churn is driven far more by *contract structure and spend behavior* (contract type, tenure, monthly and total charges) than by demographics — which means retention is something the business can actively engineer, not a fixed trait of certain customer types.

## 2. Business Problem & Objectives

Acquiring a new subscriber costs significantly more than retaining an existing one, so even a modest reduction in churn has an outsized effect on recurring revenue. Before this analysis, the business had no systematic way to answer four questions that retention spend depends on:

- Which customers are most likely to churn *right now*?
- What behavioral and contractual factors actually drive that decision?
- Which customer segments deserve which retention strategy, given limited outreach budget?
- What is the realistic revenue exposure if churn is left unaddressed?

This project was scoped to answer all four with a single, reusable analytical pipeline — one that a growth, marketing, or customer-success team could plug into their monthly reporting cycle.

### 3. Tools & Technical Stack

The project was built end-to-end in Python, chosen for the maturity of its data-science ecosystem and the ease of moving from raw CSV to a deployable model within one notebook.



| Layer                     | Tool                          | Why it was chosen                                                                                                                                                                   |
|---------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data cleaning & wrangling | Pandas                        | Fast, expressive handling of type conversion, missing values, and duplicate detection on a 7k+ row dataset.                                                                         |
| Visualization             | Matplotlib / Seaborn          | Production-quality statistical charts (distribution, box plots, stacked bars) with minimal code, ideal for stakeholder-facing reporting.                                            |
| Modeling                  | Scikit-learn<br>Random Forest | Handles nonlinear relationships and mixed feature types out of the box, resists overfitting better than a single decision tree, and outputs feature importances for explainability. |
| Evaluation                | ROC-AUC,<br>Confusion Matrix  | Accuracy alone is misleading on an imbalanced target (26% churn); ROC-AUC measures ranking quality across all thresholds.                                                           |

## 4. Dataset Overview & Data Cleaning

The dataset covers customer demographics, contract terms, subscribed services, billing amounts, tenure, and churn status for **7,043 customers**. After cleaning, **7,021 customers** remained.

### Cleaning steps and the reasoning behind each one

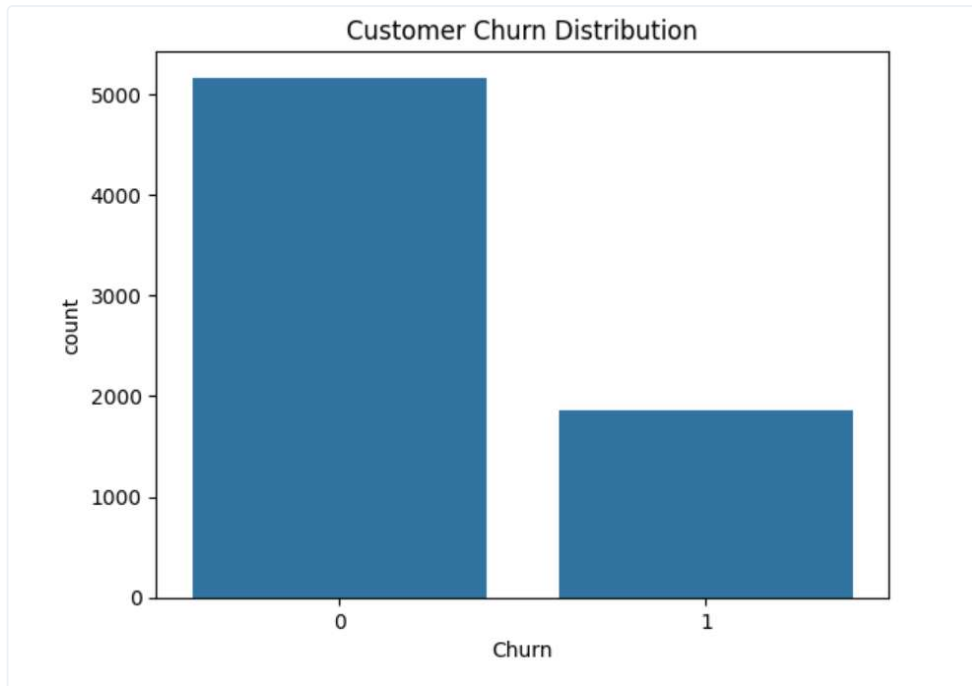
| Step                    | Action                                             | Why it mattered                                                                                                                                             |
|-------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data type correction    | TotalCharges converted from text/object to numeric | Machine learning models require numeric inputs — this column was silently stored as text and would have been unusable for modeling or correlation analysis. |
| Missing value treatment | Missing values imputed with the <b>median</b>      | Median is robust to outliers, unlike the mean — appropriate given the skew in charges data.                                                                 |
| Duplicate removal       | 22 duplicate records removed                       | Duplicate rows silently bias both descriptive statistics and model training toward over-represented customers.                                              |

## 5. Exploratory Data Analysis

Before modeling anything, the goal of EDA was to let the data show its own story — which segments churn, by how much, and why that might be. Each chart below was chosen because it isolates one specific hypothesis about churn behavior.

CHART 01

Customer Churn Distribution



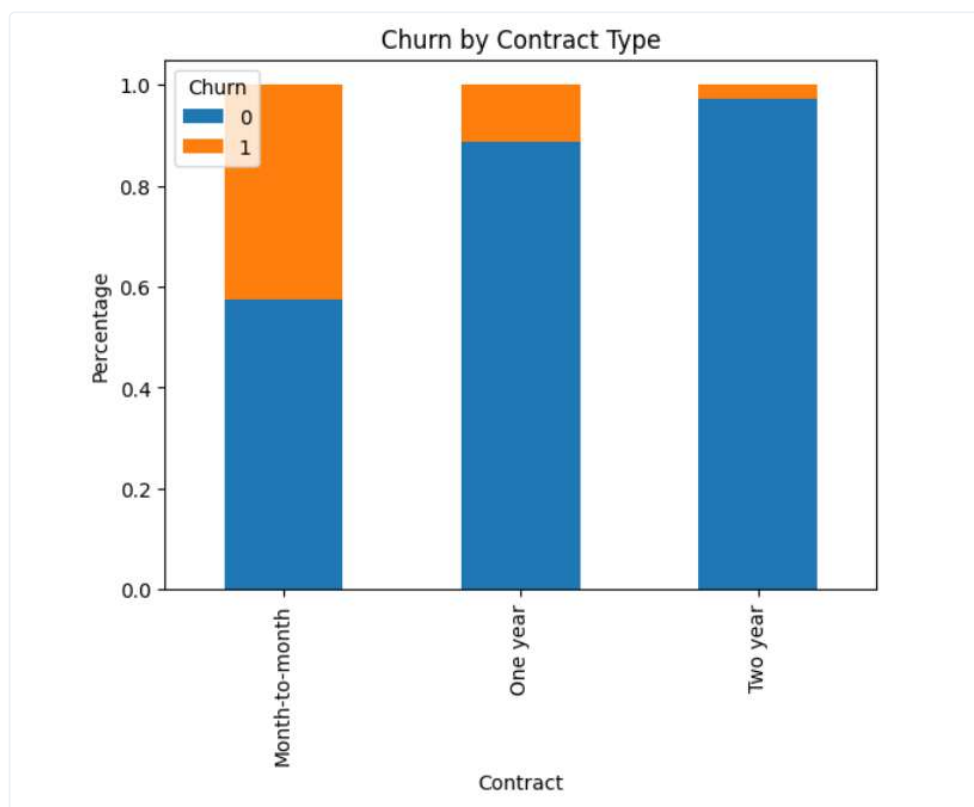
**Why this chart:** before drilling into causes, we needed a single number for the scale of the problem — what fraction of the customer base is actually at risk. A simple count plot answers that instantly.

**What it shows:** 5,164 customers were retained versus 1,857 churned — a **26.4% churn rate**, i.e. roughly one in every four customers leaves.

**Business takeaway:** a churn rate this size represents a large, addressable revenue leak — it justifies investing in a dedicated retention program rather than ad-hoc customer-service fixes.

## CHART 02

### Churn Rate by Contract Type



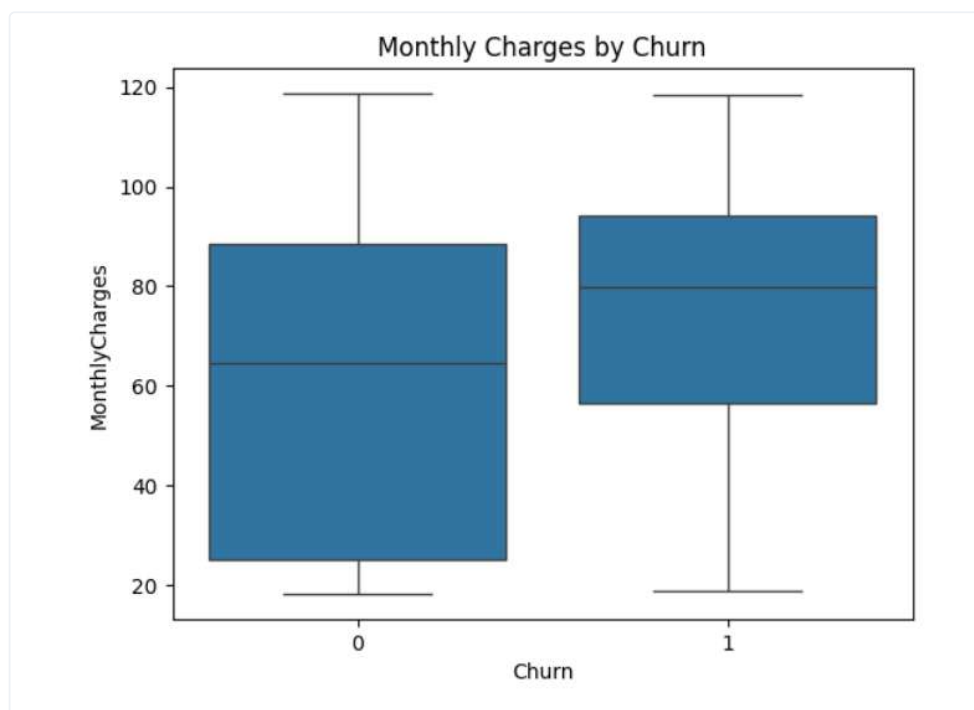
**Why this chart:** contract length is a proxy for commitment — a stacked percentage bar makes it easy to compare churn *rate* (not just raw counts) across Month-to-Month, One-Year, and Two-Year customers side by side.

**What it shows:** Month-to-Month churn sits at **42.6%**, versus **11.3%** for One-Year and just **2.8%** for Two-Year contracts — a roughly 15x difference between the least and most committed segments.

**Business takeaway:** contract type is the single strongest lever the business controls directly. Migrating even a fraction of month-to-month customers to annual plans, via incentives or bundled pricing, would materially reduce churn exposure.

### CHART 03

#### Monthly Charges by Churn Status



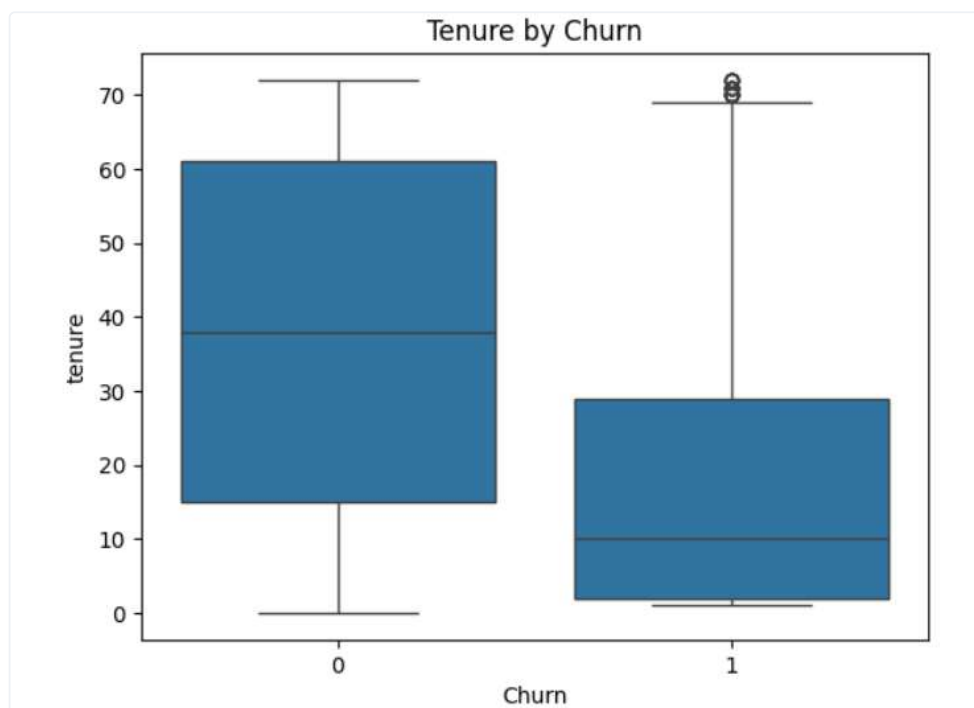
**Why this chart:** a box plot was used instead of a simple average because it also reveals spread and outliers — useful for spotting whether high spend is a consistent pattern or driven by a few extreme cases.

**What it shows:** churned customers have a noticeably higher median monthly charge (~₹74.60 average) than retained customers (~₹61.34 average) — the whole distribution for churners is shifted upward.

**Business takeaway:** price sensitivity is a real driver of attrition. Customers on higher-priced plans are candidates for proactive value reinforcement — bundling, loyalty pricing, or usage-based discounts — before they decide to leave.

#### CHART 04

##### Tenure by Churn Status



**Why this chart:** tenure is the most direct available measure of loyalty. Plotting it as a box plot split by churn status shows exactly which point in the customer lifecycle is most fragile.

**What it shows:** retained customers average **37.6 months** of tenure versus just **18.1 months** for churned customers — churn is concentrated heavily in the early relationship, not spread evenly across the customer lifetime.

**Business takeaway:** the first year is the danger zone. A structured onboarding and early-engagement program — check-ins, usage tips, milestone offers in months 1–12 — directly targets where churn actually happens.

## 6. Feature Engineering

Raw columns describe what a customer *has*; engineered features describe what they *mean* to the business. Five features were derived specifically to make the model's decisions map onto real retention actions:

| Feature      | Definition                    | Business purpose                                                                                                                     |
|--------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| AvgRevenue   | TotalCharges / (tenure + 1)   | Normalizes spend by relationship length, giving a fair "value per customer" metric regardless of how long they've been a subscriber. |
| HighValue    | Flag: spend above the average | Flags customers whose loss would disproportionately hurt revenue — the ones retention budget should protect first.                   |
| LongTenure   | Flag: tenure > 24 months      | Separates loyal, low-risk customers from the vulnerable early-tenure population identified in the EDA.                               |
| ServiceCount | Count of subscribed services  | A proxy for engagement — customers embedded in more services typically have higher switching costs.                                  |
| TenureGroup  | Tenure bucketed into ranges   | Converts a continuous number into business-readable segments (e.g. "0–12 months") for reporting and targeting.                       |

## 7. Predictive Model: Random Forest Classifier

With cleaned data and engineered features in place, a **Random Forest Classifier** was trained on an 80/20 train-test split to predict churn.

### Why Random Forest?

- **Handles nonlinear relationships** — churn is not a straight-line function of any single variable, and tree ensembles capture interaction effects (e.g. high charges *combined with* short tenure) that linear models miss.
- **Reduces overfitting** — averaging across many decision trees smooths out the noise any single tree would memorize.
- **Works natively with mixed feature types** — numeric (charges, tenure) and categorical (contract, payment method) features can be used together without heavy preprocessing.
- **Produces feature importance scores** — critical for a business report, since stakeholders need to know *why* the model flags a customer, not just that it does.

#### Confusion Matrix (Test Set)

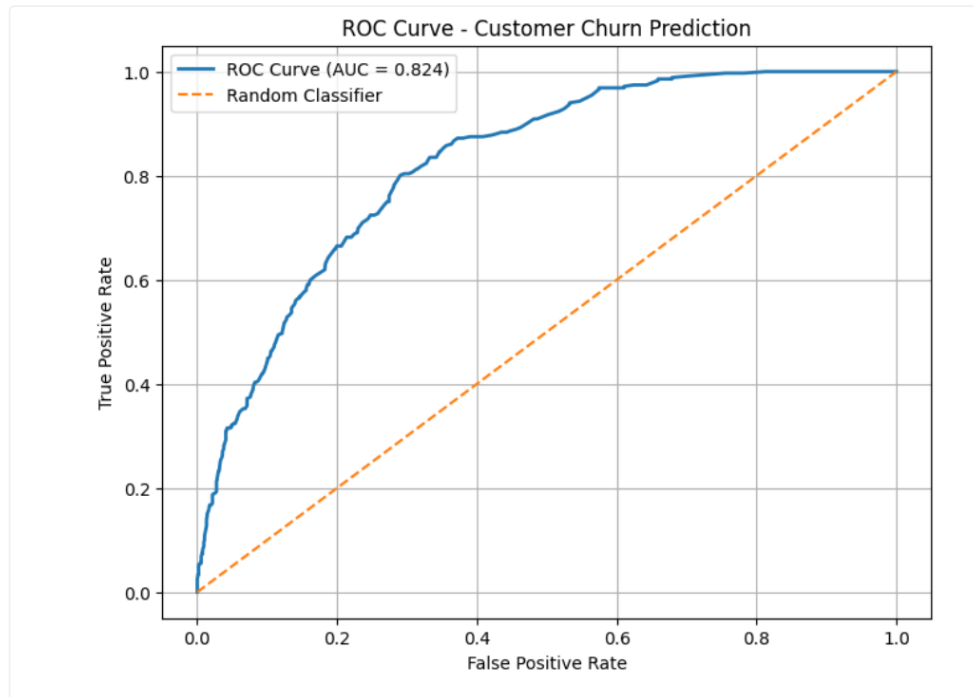
|                  | Predicted: Retained | Predicted: Churned |
|------------------|---------------------|--------------------|
| Actual: Retained | 937 (correct)       | 116                |
| Actual: Churned  | 185                 | 167 (correct)      |

**Reading it:** the model correctly identified 937 retained customers and 167 of the customers who actually churned. It missed 185 churn cases and produced 116 false alarms — a realistic trade-off given how much harder it is to predict a minority-class event like churn (26% of the base) than the majority outcome.

## 8. Model Evaluation: ROC Curve & Feature Importance

CHART 05

ROC Curve — Customer Churn Prediction



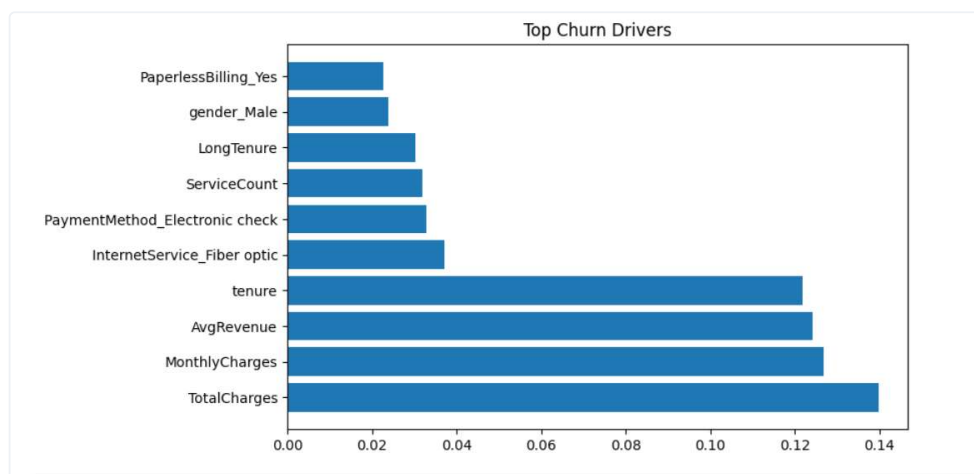
**Why this chart:** raw accuracy (78.6%) can be misleading on an imbalanced dataset, since a model that always predicts "no churn" would still be right ~74% of the time. The ROC curve instead measures how well the model *ranks* churners above non-churners across every possible decision threshold.

**What it shows:** the curve achieves an **AUC of 0.824** — meaning there is an 82.4% chance the model assigns a higher churn probability to a randomly chosen churned customer than to a randomly chosen retained one, well clear of the 0.5 "random guess" diagonal.

**Business takeaway:** an AUC above 0.8 is considered strong for churn prediction in industry practice, confirming the model's risk scores are trustworthy enough to prioritize which customers a retention team calls first.

## CHART 06

### Top Churn Drivers (Feature Importance)



**Why this chart:** a model that predicts well but can't explain itself is not actionable for a business team. Random Forest's built-in feature importance ranks every input by how much it improved the model's predictions, turning a "black box" into a business narrative.

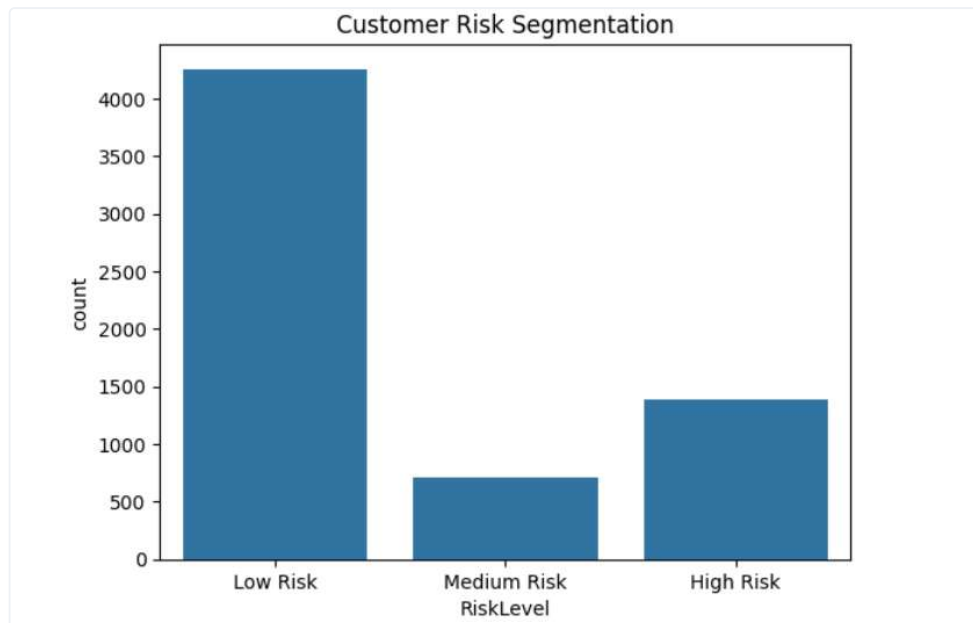
**What it shows:** **TotalCharges** (0.140), **MonthlyCharges** (0.127), **AvgRevenue** (0.124), and **Tenure** (0.122) dominate — together explaining far more of the model's decisions than services subscribed, payment method, or demographics like gender.

**Business takeaway:** retention strategy should be built around *spend and relationship length*, not demographic profiling — the model is telling us churn is a behavioral and commercial pattern, which is also the part of the relationship the business can actually influence.

## 9. Risk Segmentation & Retention Playbook

CHART 07

Customer Risk Segmentation



**Why this chart:** a single churn-probability score is hard to act on operationally. Bucketing customers into Low / Medium / High risk converts the model's output into tiers a retention or sales-ops team can directly assign budget and effort against.

**What it shows:** roughly 4,250 customers fall into Low Risk, ~715 into Medium Risk, and **1,382 customers into High Risk** — about 20% of the base needs active, near-term retention attention.

**Business takeaway:** outreach and retention budget should be allocated in that same order — High Risk customers first, Medium Risk on a lighter-touch nurture track, Low Risk left largely untouched to avoid wasted spend.

### Segment-by-segment retention playbook

| Customer Segment                  | Recommended Strategy                                         |
|-----------------------------------|--------------------------------------------------------------|
| New customers (first 12 months)   | Structured onboarding programs and early milestone check-ins |
| Month-to-month customers          | Incentivized annual-contract offers to raise switching cost  |
| High-value customers              | Loyalty rewards and dedicated account attention              |
| High monthly-charge customers     | Personalized discounts and service bundling                  |
| Model-flagged high-risk customers | Proactive outreach ahead of the point they typically churn   |

## 10. Business Impact & Conclusion

Turning this analysis into standing operations would let the business:

- Reduce churn rate by targeting the exact segments and time windows where it concentrates (early tenure, month-to-month, high-charge customers)
- Improve customer lifetime value by intervening before, not after, a customer decides to leave
- Increase recurring revenue by converting a share of month-to-month subscribers to longer commitments
- Spend marketing and retention budget more efficiently by prioritizing the 1,382 flagged high-risk accounts instead of blanket campaigns
- Run retention as a recurring, data-driven process rather than a reactive one-off effort

**In summary:** this project combined data cleaning, exploratory analysis, feature engineering, and machine learning into a single decision-support pipeline. The Random Forest model reached **78.6% accuracy** and a **0.824 ROC-AUC**, and — more importantly for the business — surfaced *why* customers churn and *which* 1,382 customers need attention now, giving a retention team a concrete, prioritized starting point rather than a guess.

*Prepared as part of a data analytics / business analytics portfolio, demonstrating an end-to-end workflow from raw data to a business-ready retention strategy.*